# NPU-Native Neural Networks on AMD XDNA 2

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**Draft** — April 5, 2026

High-throughput residual inference for HIGGS on AMD XDNA 2

**Abstract**

We study a deliberately narrow question: which neural-network shape best matches the AMD XDNA 2 NPU for high-throughput, low-latency inference on dense collider-style features? We answer it with a forward-only residual MLP whose hidden body streams across the 32 compute tiles of the Strix Point array. On the public HIGGS benchmark, we retain two key configurations: a speed-optimized $H = 32$, $L = 8$ model that sustains about 4.18M samples/s wall-clock with the tiny classifier head left on CPU, and a full-data $H = 64$, $L = 32$ model trained on 10.4M examples that reaches 77.98% test accuracy, 0.865 ROC AUC, and 0.877 PR AUC on the fixed 500k-example test split. The design is small enough to understand end to end yet strong enough to support a credible systems case for dense event filtering on consumer hardware. The contribution is a measured co-designed operating regime, not a production trigger deployment, a state-of-the-art HIGGS accuracy paper, a fully on-array inference pipeline, or an energy-efficiency study.

## 1 Introduction

Modern NPUs become most interesting when they stop being treated as "small GPUs" and start being programmed as spatial dataflow machines. Instead of launching one large shared-memory kernel, the programmer assigns a small local job to one tile, hands the activations to the next tile, and repeats until the full network has been traversed. On AMD XDNA 2, the programmer sees a 2D array of compute tiles, each with local SRAM, a VLIW plus SIMD core, a matrix-multiply unit, and explicit routing. The central question is therefore no longer just "what architecture is accurate?", but also "what architecture maps cleanly to many tiny memories connected by explicit streams?".

The focus here is deliberately narrow: a residual MLP for HIGGS, with one residual matrix per tile and a forward-only spatial-streaming deployment on the NPU. The aim is not to survey every model family or every use of XDNA 2, but to isolate one design that can be understood end to end.

HIGGS is a natural benchmark for this study. It is a dense 28-feature binary classification task derived from collider events, introduced by Baldi, Sadowski, and Whiteson as a large-scale signal-versus-background benchmark with roughly 11 million examples [6, 7]. That makes a residual MLP structurally plausible before any hardware argument is made. It also makes the throughput story easier to defend. For example, the CMS Phase-2 Level-1 trigger must inspect detector data at the 40 MHz bunch-crossing rate, produce decisions with a latency budget of about 12.5 $\mu$s, and emit accepted events at roughly 750 kHz [9]. Those rates explain why multi-megasample-per-second measurements are interesting here, but they are not used to imply deployment equivalence: the best wall-clock repeated-sample result in this paper is 4.18M samples/s, above the accepted-event rate yet far below the full bunch-crossing stream, on a simpler host-assisted prototype. The point is to study the same general systems regime of dense features, strict latency sensitivity, and pressure for very high sustained throughput.

This places the paper next to three adjacent literatures. For HIGGS itself, the reference landscape already includes both the original deep-versus-classic baseline framing and the strong tabular baselines now typically led by boosted tree ensembles [6, 13, 14, 15, 16, 17]; attention-based and decision-tree-hybrid deep tabular methods such as TabNet and NODE have also been evaluated on this benchmark [21, 22]. Comprehensive surveys of machine learning for particle physics provide broader context for where such dense-event classification sits within the HEP community [23]. For low-latency collider inference, substantial prior work has targeted FPGA-based trigger deployments and related co-design flows; the hls4ml open codesign toolchain makes the path from a trained model to firmware particularly direct [10, 19, 9]. At the accelerator level, the closest precedent is the spatial/dataflow tradition, where local SRAM and explicit movement constrain the network shape as much as raw arithmetic throughput does; purpose-built DNN accelerators and the Google TPU family both illustrate how data movement can become the primary design constraint rather than arithmetic throughput [11, 12, 20]. The claim here is narrower: given an existing benchmark, existing consumer NPU hardware, and an existing compiler/runtime stack, what is the simplest residual design that maps cleanly enough to the array to make a credible end-to-end systems case?

## 2 The hardware model we program against

Figure 1 shows the physical layout targeted here. The Ryzen AI 9 HX 370 exposes 32 compute tiles arranged as 8 columns by 4 compute rows. Below them sit 8 MemTiles, and below those 8 shim tiles that connect the array to host-visible memory and runtime control.

**Figure 1:** Physical layout of the AMD XDNA 2 NPU used here. The retained design can use all 32 compute tiles; the fastest variant uses only 2 columns, while the deeper $L = 32$ body occupies the full 8-column array.

| Property | Why it matters here |
|---|---|
| 32 compute tiles (8 columns × 4 rows) | Depth maps naturally to tile count; an $L = 8$ body uses 2 columns, an $L = 32$ body uses the full array. |
| About 64 KB local SRAM per compute tile | Parameters must stay small enough to live locally; that strongly constrains hidden width. |
| 8×8 MMUL-friendly blocked execution | Widths and batch sizes should be divisible by 8, and host-side packing must respect tiled layouts. |
| Explicit FIFOs, DMA, and routing | Data movement is part of the design, not an afterthought hidden behind a cache hierarchy. |
| IRON / MLIR-AIE / XRT software stack | The implementation programs the array close to the hardware model rather than through a high-level opaque runtime. |

**Table 1:** Only a few hardware facts matter for this paper. The most important is not the marketing TOPS number, but the combination of tiny tile memories and explicit data movement.

These constraints shape the model directly. The hidden body is kept square so that each tile stores one $H \times H$ residual weight matrix. Widths are kept as multiples of 8 to match the AIE MMUL datapath. The host only moves the data that are still cheaper to keep on CPU today: the input projection and, for the fastest retained variant, the tiny final $H \to 2$ classifier head.

## 3 A residual network that matches the array

The network studied here is intentionally minimal.

### Notation

We use $H$ for hidden width, $L$ for the number of residual blocks, and $B$ for batch size. In the deployment path studied here, one residual block maps to one compute tile. We use $\to$ for layer mappings (for example $H \to 2$ classifier) and $\times$ for matrix shapes (for example $H \times H$ weights).

### Model definition

$$h_{\ell+1} = \mathrm{ReLU}(h_\ell W_\ell) + h_\ell, \qquad \ell = 0, \ldots, L-1.$$

The skip connection follows the convention introduced by He et al. [18]; each residual matrix $W_\ell$ is assigned to one compute tile. Activations enter the array, flow through the tiles in a snake, and return to the host only after the full hidden body has completed. In the current deployment path:

- the host CPU runs the input projection from the native 28 HIGGS features to hidden width $H$,
- the NPU runs the residual body,
- the host CPU currently keeps the tiny final $H \to 2$ head when that is faster than the available fused on-array tail.

This design leads to two retained operating variants.

- **Speed-optimized variant:** $H = 32$, $L = 8$, mapped to 2 columns. This is the fastest end-to-end HIGGS point measured so far.
- **Accuracy-tuned variant:** $H = 64$, $L = 32$, trained on the full public dataset and mapped to the full 8-column array when the deeper body is streamed on the NPU.

The retained paper path keeps the narrow surround on CPU and uses the NPU only for the residual streaming body. The same snake-order placement then scales from the 2-column $L = 8$ speed-optimized variant to the full 8-column $L = 32$ body; Figure 2 makes that deployment contract concrete.

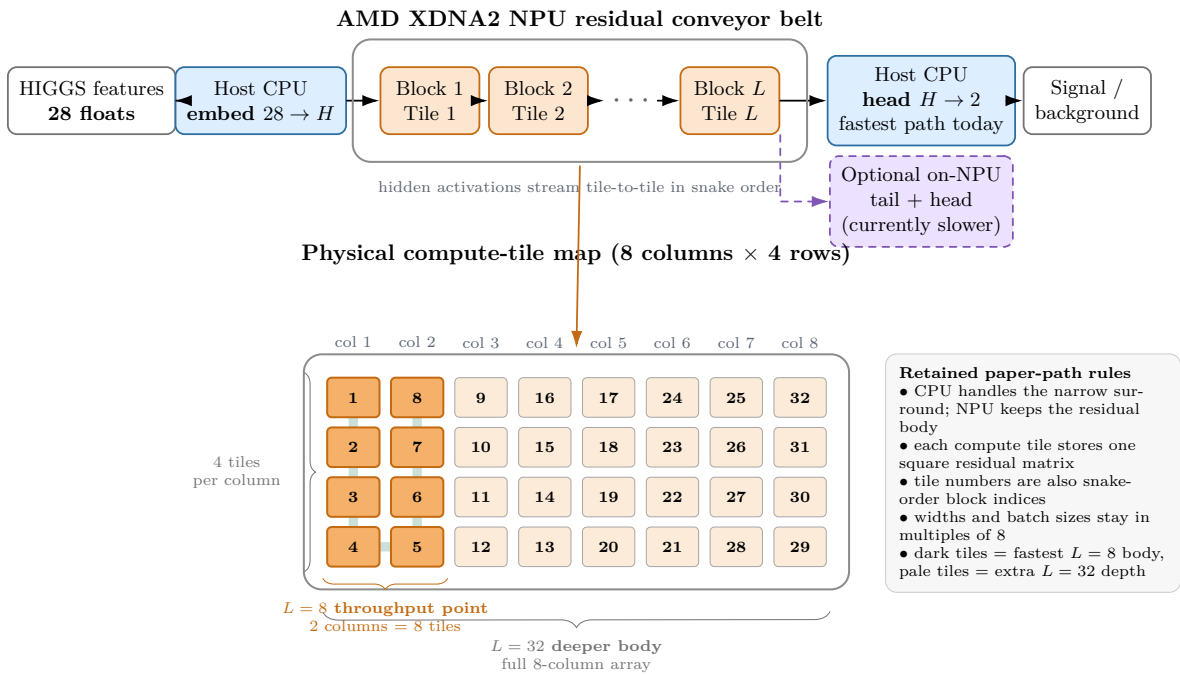


**Figure 2:** Model-to-hardware mapping for the retained HIGGS/XDNA 2 deployment path. The host CPU keeps the cheap $28 \to H$ input projection and, in the fastest retained path, the tiny $H \to 2$ head. The NPU assigns one $H \times H$ residual block per compute tile, using 2 columns for the $H = 32$, $L = 8$ speed-optimized variant and the full 8-column array for the deeper $L = 32$ body.

The important design choice is not just "use an MLP", but "use an MLP whose depth and width map directly to the array's tile structure". That keeps the design simple enough to analyze and direct enough to deploy.

# 4 From Python model to NPU artifact

The toolchain is deliberately short. `aie_kernels/matmul_relu_skip.cc` defines the tile-local residual kernel. `resmlp/streaming_design.py` turns a chosen $(H, B, L)$ shape into a spatial-streaming MLIR-AIE program. `resmlp/streaming_op.py` wraps the compiled XCLBIN and instruction binary as a resident XRT operator. Finally, `resmlp/streaming_infer.py` drives trained checkpoints through that operator and measures wall-clock throughput.

The novelty claimed here is not a new compiler or runtime. IRON and MLIR-AIE already provide the compilation substrate [1, 2]; what is new is the model-to-array mapping, the host/NPU partition, and the end-to-end measurements of that co-designed path on real HIGGS checkpoints.

A key detail of the current forward path is that the residual weights are embedded at compile time. Each throughput run therefore builds a checkpoint-specific streaming artifact. That is not the most general deployment contract. It is, however, the cleanest way to study the geometry, routing, and wall-clock behavior of a resident spatial-streaming design.

# 5 Training protocol on HIGGS

The data pipeline uses the public `jxie/higgs` mirror of the published train/test splits [8]. The local cache materializer produces a fixed 10.5M-example train split and a fixed 500k-example test split. For model selection, training holds out 100k examples from the train split. That leaves 10.4M examples for optimization. All reported runs use the native 28-dimensional feature vector without any padded-width preprocessing. Hyperparameter search and checkpoint selection use

| Model | Training recipe |
|---|---|
| $H = 32$, $L = 8$ full-data pilot | 5 epochs on the AMD Radeon 890M ROCm path; AdamW, batch 8192, cosine $3 \times 10^{-3} \rightarrow 10^{-4}$, weight decay $10^{-4}$, no label smoothing. |
| $H = 32$, $L = 32$ long schedule | 20 epochs on the AMD Radeon 890M ROCm path; AdamW, batch 8192, cosine $3 \times 10^{-3} \rightarrow 10^{-4}$, weight decay $10^{-4}$, no label smoothing. |
| $H = 64$, $L = 32$ validation-selected tuning run | Best checkpoint at epoch 46 of a 50-epoch MLflow/Optuna sweep on the AMD Radeon 890M ROCm path; AdamW, batch 4096, cosine $1.069 \times 10^{-3} \rightarrow 8.70 \times 10^{-5}$, weight decay $3.23 \times 10^{-3}$, label smoothing 0.0197. |

**Table 2:** Training recipes for the reported HIGGS checkpoints. Both direct runs and validation-driven hyperparameter search were used to produce the reported models.

held-out validation accuracy. The 500k-example test split is used only for final evaluation of already-selected models.

HIGGS is also a good benchmark because it sits at the intersection of two real literatures. In high-energy physics it is a canonical dense-event benchmark [6], situated within a broader programme of applying machine learning to particle-physics classification and reconstruction that is surveyed in [23]. In tabular machine learning it is one of the standard large-data problems on which tree ensembles such as XGBoost, LightGBM, and CatBoost remain strong baselines, while modern deep tabular models—including attention-based and decision-tree-hybrid approaches—are now more competitive but not universally dominant [13, 14, 15, 16, 17, 21, 22]. That means the relevant bar is not "beat every HIGGS leaderboard", but "land in a statistically credible part of the baseline landscape while preserving a deployment shape the NPU can actually exploit". Good-enough accuracy on HIGGS is therefore genuinely worth caring about even if the main contribution of this work remains systems-oriented.

# 6 Results

## 6.1 Accuracy on the full public HIGGS split

Table 3 summarizes the current full-data results. The key point is that the systems design is not accuracy-indifferent. Even this deliberately hardware-friendly residual MLP becomes a credible classifier once it is trained on the full public HIGGS data.

Two key findings emerge. First, depth helps. Moving from $L = 8$ to $L = 32$ improves every metric even before any serious hyperparameter search. Second, validation-driven hyperparameter tuning lifts the result from the high-0.85 ROC band to 0.865 ROC AUC while also improving accuracy, PR AUC, and log loss. This is still not a claim of state-of-the-art HIGGS accuracy, but it is strong enough that the main contribution of the paper is no longer accuracy-indifferent.

## 6.2 Throughput of the spatial-streaming deployment

Table 4 reports the current repeated-sample throughput measurements over 50M processed samples. "Wall throughput" includes the host input projection, host head, host-visible runtime overheads, and the NPU residual stage. "Kernel throughput" counts only the NPU kernel wait time.

This table sharpens the systems conclusion. The best current wall-clock point is not the deepest model, and it is not the version with the entire graph moved onto the array. The tuned $H = 64$, $L = 32$ checkpoint now runs cleanly on hardware, but it lands at only 1.35M samples/s wall-clock, so the additional width extends the accuracy frontier rather than the throughput frontier. On the width extends the accuracy frontier rather than the throughput frontier.

| Model | Test acc. | ROC AUC | PR AUC | Log loss |
|---|---|---|---|---|
| $H = 32$, $L = 8$ full-data (5 epochs) | 75.89% | 0.8430 | 0.8559 | 0.4861 |
| $H = 32$, $L = 32$ full-data (5 epochs) | 76.56% | 0.8502 | 0.8628 | 0.4762 |
| $H = 32$, $L = 32$ full-data (20 epochs) | 76.98% | 0.8542 | 0.8668 | 0.4705 |
| Validation-selected tuning run: $H = 64$, $L = 32$ | 77.98% | 0.8653 | 0.8770 | 0.4543 |

**Table 3:** Full-data HIGGS results on the public `jxie/higgs` mirror. Validation-selected tuning at $H = 64$, $L = 32$ reaches 0.865 ROC AUC while remaining within the same residual design family.

| Configuration | Head placement | Wall throughput | Kernel throughput |
|---|---|---|---|
| $H = 32$, $L = 8$, 2 columns | CPU head | 4.18M samples/s | 26.29M samples/s |
| $H = 32$, $L = 8$, 2 columns | NPU head | 3.65M samples/s | 7.47M samples/s |
| $H = 32$, $L = 32$, 8 columns | CPU head | 3.49M samples/s | 19.06M samples/s |
| $H = 32$, $L = 32$, 8 columns | NPU head | 3.15M samples/s | 6.68M samples/s |
| $H = 64$, $L = 32$, 8 columns | CPU head | 1.35M samples/s | 3.05M samples/s |

**Table 4:** Repeated-sample throughput on the retained HIGGS checkpoints. The fastest wall-clock path is the 8-layer, 2-column variant with the classifier head left on CPU; the current fused on-array head is slower.

On the present tail kernel, the fused $H \to 2$ NPU head is slower than keeping that tiny classifier on CPU. That is an important honest result: the fastest measured path is not yet a fully on-array pipeline, and the right abstraction boundary in the current implementation is a CPU input projection, NPU residual body, and CPU head. The next hardware step is therefore not "force everything on-array" but "move the surrounding stages only when they actually improve wall time". A plausible reason is that the $H \to 2$ head is simply too small to amortize the extra synchronization, buffer traffic, and launch overheads of the fused tail path. In other words, the added runtime machinery costs more than the CPU work it removes.

Figure 3 combines the retained full-data accuracy rows with the measured CPU-head wall-throughput rows. It makes the trade-off explicit: $H = 32$, $L = 8$ is the wall-clock leader, $H = 32$, $L = 32$ trades some speed for better accuracy, and the tuned $H = 64$, $L = 32$ run remains the accuracy ceiling, but at 1.35M samples/s wall-clock it sits well left of the two $H = 32$ variants.

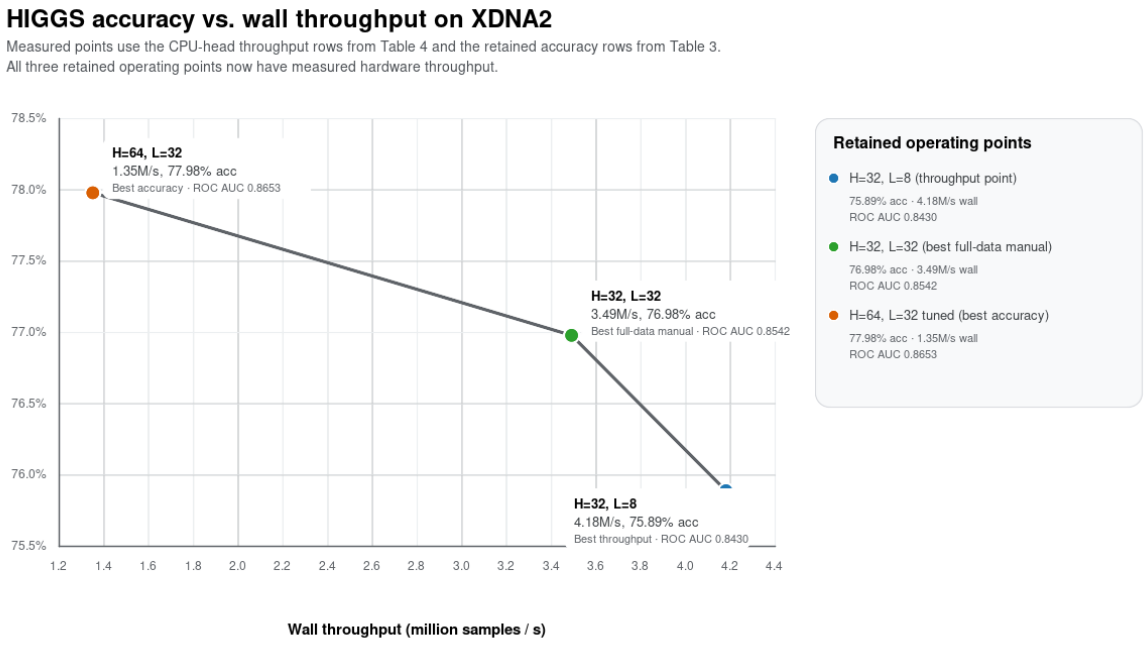


**Figure 3:** Accuracy-throughput frontier for the retained HIGGS/XDNA 2 variants. X-axis: test accuracy; Y-axis: wall-clock throughput in M samples/s. The three plotted throughput measurements use the CPU-head rows from Table 4, including the tuned $H = 64$, $L = 32$ checkpoint.

# 7 Discussion

The claim is deliberately narrow but defensible.

- A residual MLP can be shaped to match the AMD XDNA 2 array very directly: one residual matrix per tile, streamed activations, explicit routing, and checkpoint-specific resident weights.
- That design already delivers multi-million-sample-per-second inference on a real collider-style dense benchmark using cheap consumer hardware.
- The same design family also reaches nontrivial full-data HIGGS quality, culminating in 77.98% test accuracy and 0.865 ROC AUC on the fixed 500k-example test set.

Just as importantly, the paper avoids claims it does not need. It does not present the current model as a production trigger deployment, the current HIGGS result as state of the art, the fastest path as a fully on-array pipeline, or the present measurements as an energy-efficiency study. The benchmark is public, the hardware is existing AMD XDNA 2, and the compiler/runtime substrate is pre-existing; none of those ingredients are introduced here. The novelty is the co-designed operating regime: a residual MLP whose hidden-body geometry matches the array closely enough that explicit spatial deployment yields both a useful throughput result and a statistically credible HIGGS checkpoint.

A useful boundary from the SRAM accounting is that wider hidden bodies and wider surrounding inputs do not stress the current deployment in the same way. For the present $B = 8$ spatial-streaming layout, an internal residual tile carries roughly $1024 + 2H^2 + 32H$ bytes, while the ingress tile carries $1024 + 2H^2 + 64H$ because it must hold both the host-ingress buffers and the first residual stage. That makes the first tile, not the middle of the belt, the width bottleneck. The retained layout still fits at $H = 160$ with about 62.5 KB live on the ingress tile, but the same accounting already pushes $H = 168$ past 64 KB there even though the internal tiles would still fit. In the current paper, then, widening the hidden body is a genuine tile-SRAM limit of the current spatial-streaming placement.

The surrounding stages are different. For the native 28-feature HIGGS input, even a

padded $32 \to H$ input projection carries only $64H$ bytes of bf16 weights (4 KB at $H = 64$, 10 KB at $H = 160$), so keeping that stage on CPU today should not be read as evidence of an SRAM bottleneck. The awkward part is that 28 inputs do not naturally match the $8\times8$ MMUL granularity, so an on-array ingress path must pay packing, alignment, and boundary-management costs. The same distinction is even clearer for the tiny $H \to 2$ head: its weight footprint is negligible, and the measured slowdown of the fused NPU head is best understood as synchronization and launch overhead. This keeps the scope narrow. Within the HIGGS-style feature regime studied here, wider surrounding inputs mainly stress alignment and runtime boundaries, whereas the hard SRAM wall comes from widening the hidden body under the present first-tile layout.

## 8 Conclusion and next steps

The main result is simple: a network designed for spatial compute on AMD XDNA 2 can deliver unusually high throughput on real dense-event classification while remaining small enough to understand and validate end to end. The model family is deliberately modest, but it maps cleanly to the array and yields both a strong speed-optimized variant and a nontrivial full-data HIGGS result.

What exists today is a measured host-assisted prototype; fully on-array ingress/egress and energy-efficiency characterization remain future work.

The next steps are correspondingly focused. The highest-leverage systems step is still to move the $28 \to 32$ input projection on-array so that the surrounding host work shrinks further, but that should be understood mainly as an alignment/runtime clean-up rather than as a fix for a tile-SRAM shortfall. The SRAM-sensitive scaling question is different: wider hidden bodies will require revisiting the first-tile ingress layout because that tile reaches the 64 KB ceiling before the internal tiles do. The highest-leverage modeling step is to continue exploiting the promising $H = 64$, $L = 32$ region with longer schedules and regularization, rather than broadening the scope of the architecture prematurely.

# A AMD XDNA 2 background

The main text intentionally keeps only the hardware facts that directly matter for the systems argument. For readers less familiar with AI Engine-class NPUs, Figure 4 gives a programmer's logical view of one compute tile. It is not a physical floorplan. What matters here is that each tile combines local SRAM, a VLIW+SIMD control core, a tile-local MMUL datapath, and explicit DMA / FIFO connectivity. That combination makes weight-stationary streamed mappings natural on this hardware: weights can stay local, activations arrive through explicit buffers, and data movement must be planned rather than delegated to a large shared cache hierarchy [4, 3].

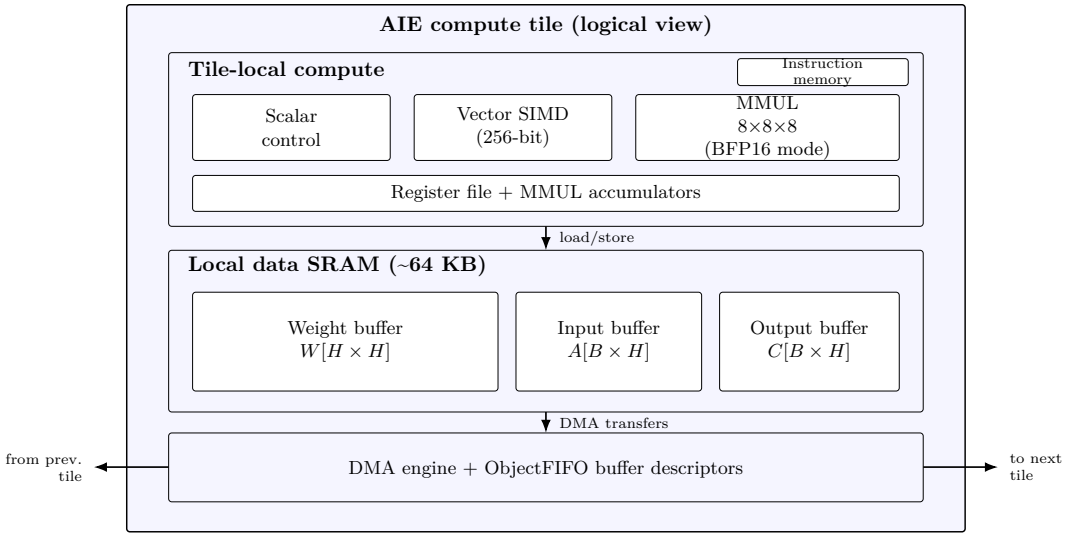


**Figure 4:** Logical view of one XDNA 2 compute tile from a programmer's point of view. The figure is schematic rather than a physical floorplan. What matters for this paper is the combination of tile-local SRAM, VLIW+SIMD control, MMUL-backed matrix math, and explicit DMA / FIFO movement between tiles.

Table 5 collects the hardware terms that recur most often in the paper. It is meant as a compact glossary, not an exhaustive one, but it gives enough background to follow the deployment discussion without already knowing the AI Engine vocabulary.

| Term | Meaning in this paper |
|---|---|
| **AIE tile** | One compute tile of the XDNA 2 array: local SRAM, local compute, instruction memory, and DMA connectivity. |
| **MemTile** | Memory-only tile between shim tiles and compute rows; useful for routing and on-chip buffering. |
| **Shim tile** | Interface tile that connects the array to host-visible memory and runtime infrastructure. |
| **DMA** | Hardware transfer mechanism used to move data between host-visible buffers and tile-local SRAM without CPU-managed copy loops. |
| **SIMD** | Single Instruction, Multiple Data. One instruction operates on a vector of values at once. |
| **VLIW** | Very Long Instruction Word. One instruction can encode multiple parallel operations such as loads, arithmetic, and control. |
| **MMUL** | Tile-local matrix-multiply datapath used by the AIE kernel intrinsics; it strongly favors blocked 8×8 layouts. |
| **ObjectFIFO** | IRON abstraction for explicit producer/consumer channels between tiles and host boundary stages. |
| **XCLBIN / XRT** | The compiled device artifact and the runtime that loads and launches it on the NPU. |

**Table 5:** Compact glossary for the AMD XDNA 2 terms that are used repeatedly in the main text.

# B HIGGS physics background

The HIGGS benchmark comes from a standard collider-analysis setting: a rare target process must be separated from much more common look-alike processes using reconstructed event kinematics [6]. The label is binary. In the UCI release, '1' denotes signal and '0' denotes background [7]. Physically, that does not mean the detector has directly "seen a Higgs." It means the reconstructed pattern of visible objects in the event is more consistent with the target signal process than with ordinary background production.

Each row is therefore already one reconstructed collision event, not raw detector data. The low-level variables describe the visible ingredients of the event: one charged lepton, missing transverse momentum, and up to four jets, plus per-jet b-tag indicators [7]. In collider language, $p_T$ is momentum transverse to the beam, $\eta$ and $\phi$ encode direction, and missing transverse momentum records the momentum imbalance associated with invisible particles leaving the detector. The high-level variables are invariant-mass combinations such as $m_{jj}$, $m_{jjj}$, $m_{lv}$, $m_{jlv}$, $m_{bb}$, $m_{wbb}$, and $m_{wwbb}$; they test whether groups of reconstructed objects line up with a plausible underlying resonance or decay chain [7].

That structure is exactly why HIGGS is useful here. The model does not have to solve raw reconstruction. It receives a compact physics summary of each event and must make a fast signal-versus-background decision on dense numeric inputs. In this repository those examples come from the public `jxie/higgs` mirror of the published train/test split [8].

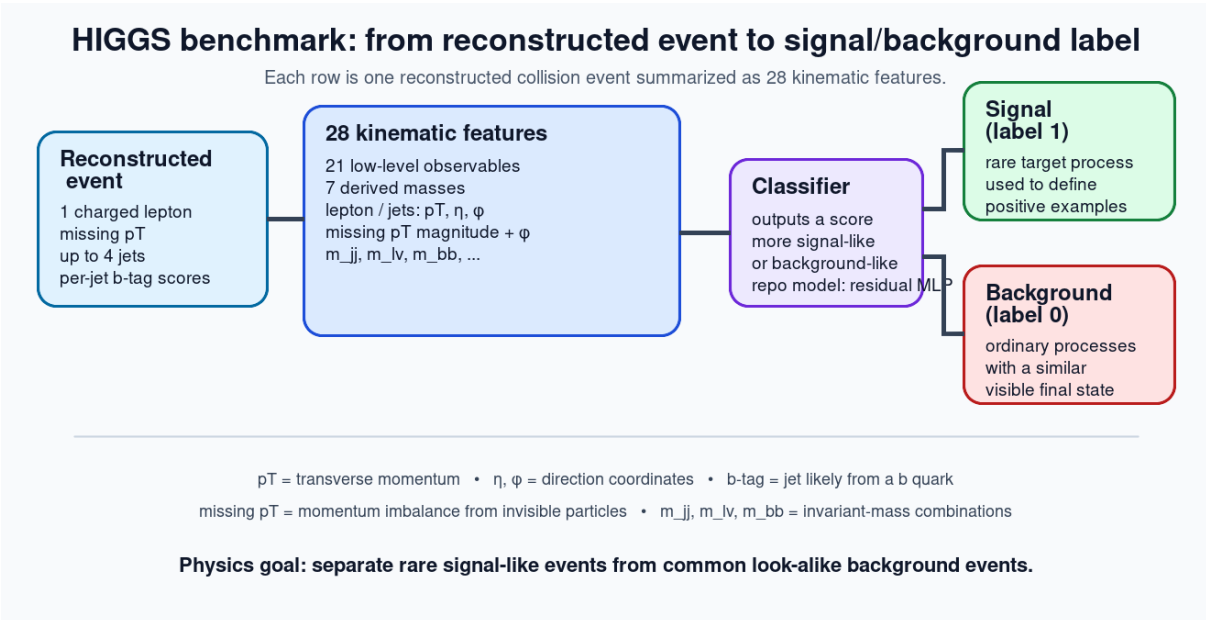


**Figure 5:** From reconstructed collision event to binary label in HIGGS. The dataset stores lepton, jet, missing-momentum, and derived invariant-mass features; the classifier must separate rare signal-like events from common look-alike background events.

## Disclosure of AI-assisted workflow

AI coding and research assistants, including Claude Opus 4.6, GPT-5.4, and Gemini 3.1 Pro, were used extensively during this project to generate code, propose experiments, summarize results, review literature, and critique or draft text. The author curated the research direction, selected the final claims and figures, verified the reported results, and takes responsibility for the content of this paper.

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